

Lecture 9/Module 1

DC Circuit Analysis and Theorems

05/08/2026

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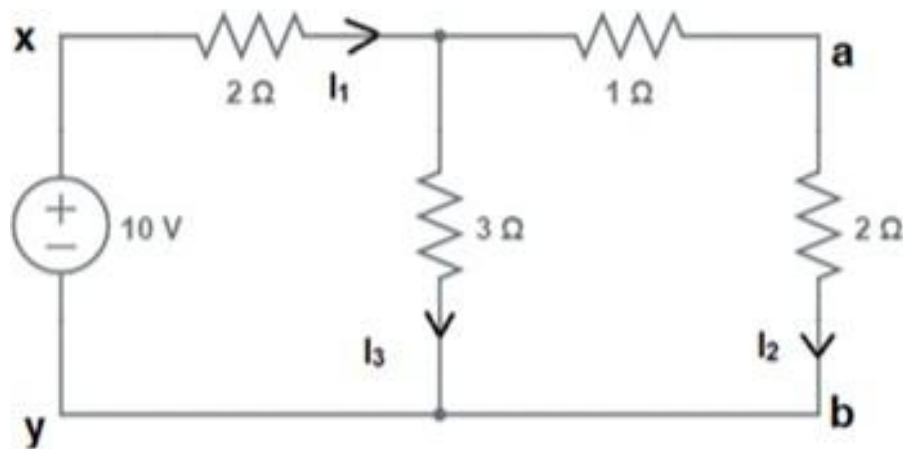
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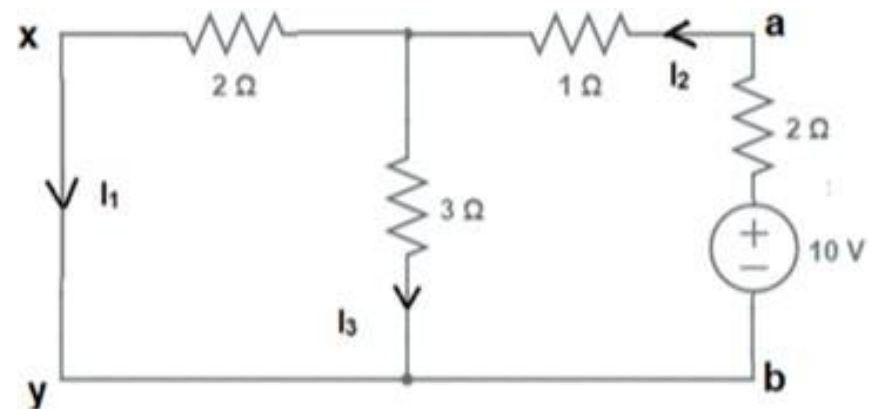
In a linear, bilateral network containing only one independent source, if a voltage source V applied in branch A produces a current I in branch B, then placing the same voltage source V in branch B will produce the same current I in branch A.

If a current source I placed in branch A produces a voltage V across branch B, then placing the same current source I in branch B will produce the same voltage V across branch A.



$$I_1 = 1.4 \text{ A}$$

$$I_2 = 1.4 \text{ A}$$



Reciprocity Theorem

The limitation of this theorem:

It is applicable only to single-source networks and not in the multi-source network.

The network where reciprocity theorem is applied should be linear and consist of resistors, inductors, capacitors and coupled circuits.

The circuit should not have any time-varying elements.

Reciprocity Theorem

Steps for solving a network utilizing Reciprocity Theorem:

Step 1 – Firstly, select the branches between which reciprocity has to be established.

Step 2 – The current in the branch is obtained using any conventional network analysis method.

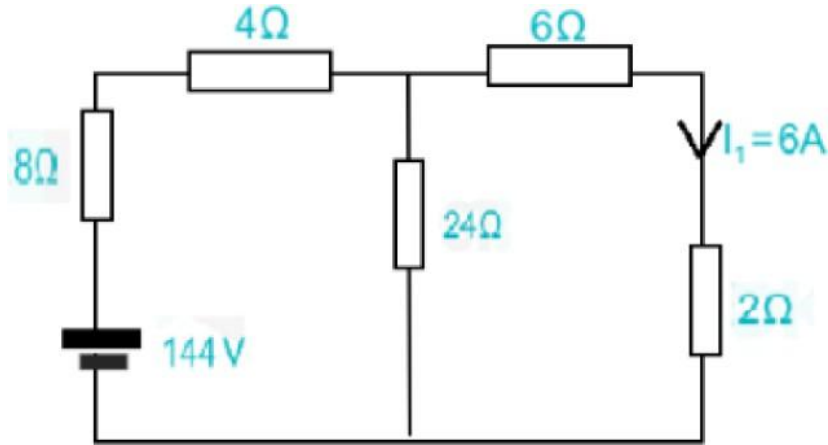
Step 3 – The voltage source is interchanged between the branch which is selected.

Step 4 – The current in the branch where the voltage source was existing earlier is calculated.

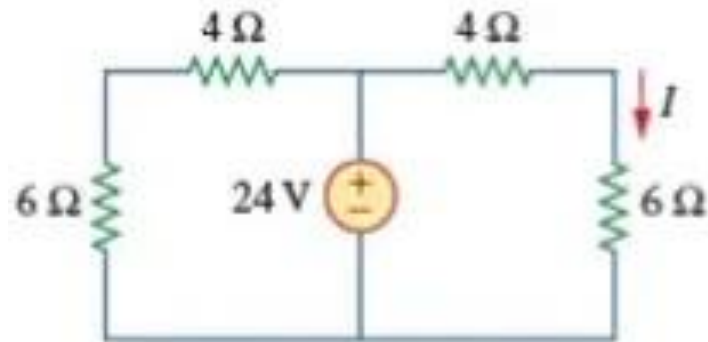
Step 5 – Now, it is seen that the current obtained in the previous connection, i.e., in step 2 and the current which is calculated when the source is interchanged, i.e., in step 4 are identical to each other.

Reciprocity Theorem

Validate the reciprocity theorem.



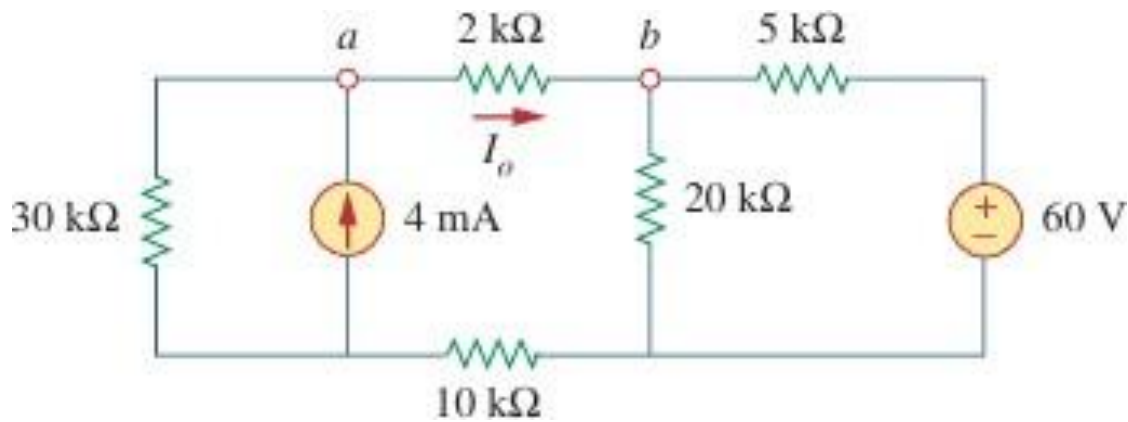
(a)



(b)

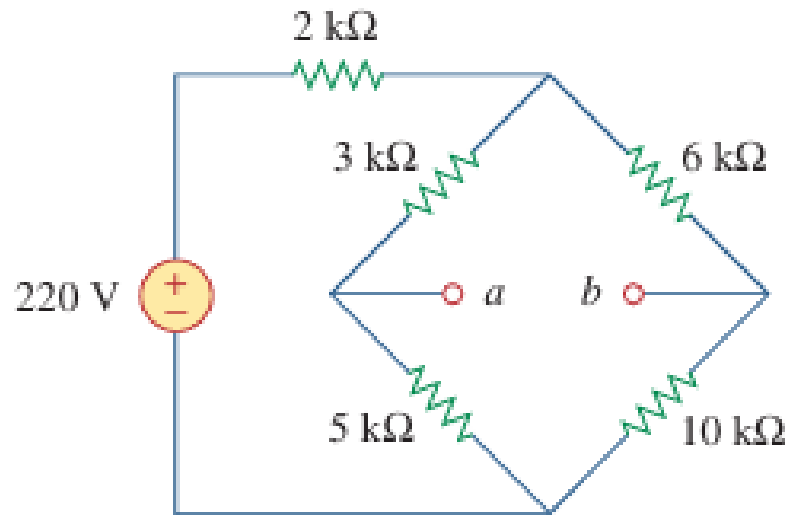
Exercise

Consider the circuit given below. An ammeter with internal resistance is inserted between A and B to measure I_o . Determine the reading of the ammeter if: (a) $R_i = 500\ \Omega$ (b) $R_i = 0\ \Omega$ (Hint: Find the Thevenin's equivalent circuit at terminals.

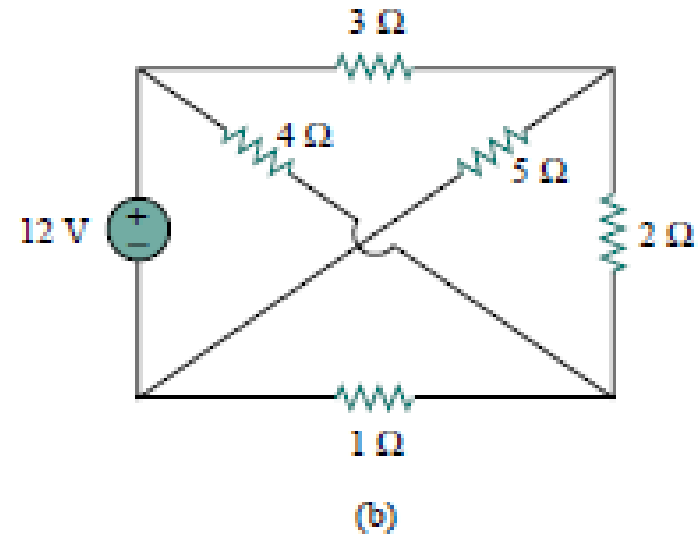
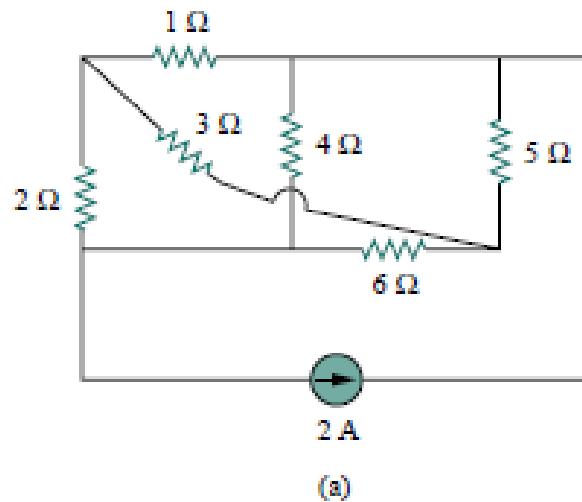


Exercise

Consider the bridge circuit below. Is the bridge balanced? If the 10-k resistor is replaced by an 18-k resistor, what resistor connected between terminals absorbs the maximum power? What is this power?

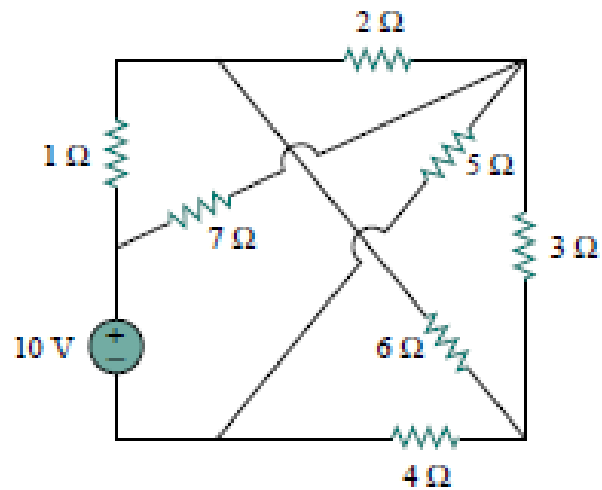


Planar and Non-Planar Circuit

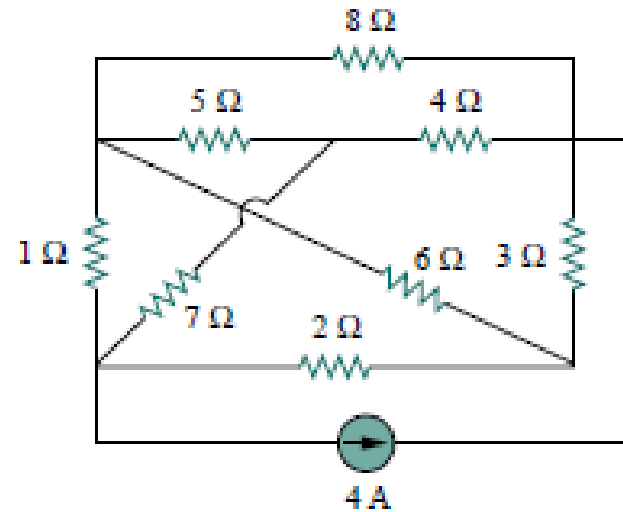


Note: Mesh Analysis we don't do on non-planar circuit in standard form

Planar or Non-Planar ??

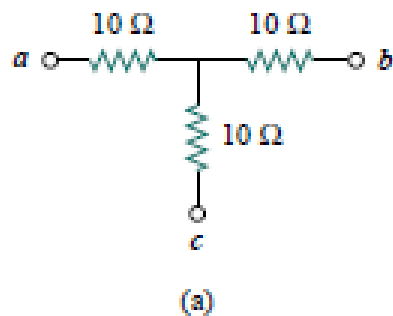


(a)



(b)

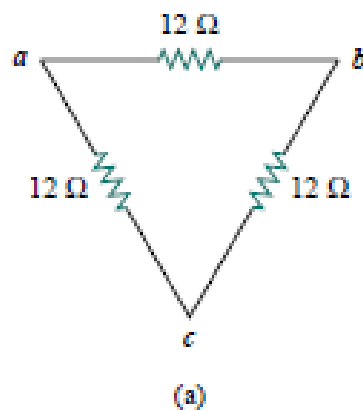
2.43 Convert the circuits in Fig. 2.107 from Y to Δ .



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Figure 2.107 For Prob. 2.43.

2.44 Transform the circuits in Fig. 2.108 from Δ to Y.



??

Figure 2.108 For Prob. 2.44.

Lecture 9/Module 2

Introduction, DC Circuit Analysis and Theorems

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First order circuits

Applying Kirchhoff's laws to purely resistive circuits results in algebraic equations, while applying the laws to RC and RL circuits produces differential equations.

The differential equations resulting from analyzing RC and RL circuits are of the first order.

A first-order circuit is characterized by a first-order differential equation.

There are two types of first-order circuits (RC and RL), and there are two ways to excite the circuits.

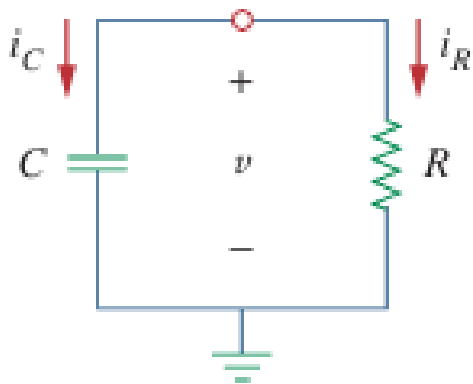
The first way is by initial conditions of the storage elements in the circuits. In these so-called source-free circuits, we assume that energy is initially stored in the capacitive or inductive element. The energy causes current to flow in the circuit and is gradually dissipated in the resistors.

The second way of exciting first-order circuits is by independent sources.

The Source-Free RC Circuit

A source-free RC circuit occurs when its dc source is suddenly disconnected.

The energy already stored in the capacitor is released to the resistors.



Consider a series combination of a resistor and an initially charged capacitor.

Since the capacitor is initially charged, we can assume that at time $t = 0$, the initial voltage is not zero but,

$$v(0) = V_0$$

with the corresponding value of the energy stored as

$$w(0) = \frac{1}{2}CV_0^2$$

Thank You